

Ex. No: 9Word2Vec Embedding Model with 2D Visualization

Implement Word2Vec Embedding model for any small and large corpus datasets and Visualize 2D Embeddings in terms of Dimensionality Reduction Techniques: PCA, t-SNE, and UMAP

Datasets:

- 1. Any Small Corpus
- 2. Any Large Corpus

Aim

To implement a Word2Vec embedding model (both CBOW and Skip-gram variants) for small and large corpus datasets and to visualize the learned high-dimensional word embeddings in 2D space using Principal Component Analysis (PCA), t-Stochastic Neighboring Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP).

Procedure/Pseudocode

- 1. Load and preprocess small corpus (e.g., text8) and large corpus (e.g., Google News / Wikipedia dump).
- 2. Train Word2Vec model using Gensim with parameters: vector_size=100/300, window=5, min_count=5, epochs=10.
- 3. Train both CBOW (sg=0) and Skip-gram (sg=1) variants.
- 4. Extract embedding matrix: each row = embedding vector for a vocabulary word.
- 5. PCA: Apply sklearn's PCA to reduce embedding dimension to 2 → scatter plot with word labels.
- 6. t-SNE: Apply TSNE with perplexity=30, n_iter=1000 → scatter plot showing local cluster structure.
- 7. UMAP: Apply UMAP with n_neighbors=15, min_dist=0.1 → scatter plot showing global and local structure.
- 8. Color-code points by semantic category (animals, countries, verbs, etc.) for visual interpretation.

Libraries Used

- 1. gensim – for Word2Vec model training
- 2. sklearn.decomposition.PCA – for PCA dimensionality reduction
- 3. sklearn.manifold.TSNE – for t-SNE reduction
- 4. umap-learn – for UMAP reduction
- 5. matplotlib – for 2D scatter plot visualization
- 6. numpy – for vector operations

Test Cases

- 1. Word2Vec analogy: king - man + woman $\approx$ queen (verified using `most_similar()`)
- 2. PCA plot: Country names (France, Germany, Japan) cluster together in 2D space
- 3. t-SNE plot: Animal names form distinct cluster separate from country names
- 4. UMAP plot: Preserves both local clusters and global topology better than t-SNE

Results

Word2Vec embeddings captured rich semantic and syntactic relationships. PCA revealed major variance directions but lost local structure. t-SNE produced clear semantic clusters for related words. UMAP provided the best balance between local cluster preservation and global structure. Skip-gram embeddings showed superior quality for rare words in large corpus experiments.